

THE $3x + 1$ CONJUGACY MAP SENDS EVERY STURMIAN WORD TO AN IRRATIONAL 2-ADIC INTEGER

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ABSTRACT. Let T be the $3x + 1$ map on the 2-adic integers and Φ the Bernstein–Lagarias conjugacy map, sending a parity vector to its realising 2-adic integer; for $s \in \{0, 1\}^{\mathbb{N}}$ let $p_s(L)$ count its distinct length- L factors. Dubickas proved in 2009 that a positive-integer $3x + 1$ trajectory tending to infinity has parity-word complexity $p_s(L) > 1.70951129 L$ eventually. His argument transports to every rational 2-adic integer, since T preserves a fixed odd denominator: if s is not eventually periodic and $\Phi(s) \in \mathbb{Q}$, then $\liminf_L p_s(L)/L \geq 1/\log_2(3/2) = 1.70951\dots$. Since every Sturmian word has $p_s(L) = L + 1$, this gives $\Phi(s) \notin \mathbb{Q}$ for every Sturmian word, with margin $2 - \log_2 3$.

We give a second, independent and effective proof. Let $\beta = \ln 2 / \ln 3$ and $1c_\beta$ the critical-density Sturmian word of López and Stoll, the hardest case here: periodic words at the convergents p_n/q_n of β agree with $\Phi(1c_\beta)$ to 2-adic depth exactly $q_n + q_{n+1} - 1$ from above, and a Liouville comparison closes the argument using only $q_n \rightarrow \infty$. This extends to every irrational slope, and a further argument removes the restriction to intercept 0 via an initial square of the word itself.

New here: the transport to rational 2-adic integers and its Sturmian corollary, settling López–Stoll’s 2009 question at this slope; the exact depth law; the height floors; and the independent effective Liouville proof. Counting reaches every complexity below $1.70951\dots L$; the frontier is complexity $2L$ (Rote words, rotation codings), open for both routes.

1. INTRODUCTION

Let $T: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ be the $3x + 1$ map in its standard form,

$$T(x) = \begin{cases} x/2, & x \equiv 0 \pmod{2}, \\ (3x + 1)/2, & x \equiv 1 \pmod{2}. \end{cases}$$

The *parity vector* of $x \in \mathbb{Z}_2$ is the sequence $v(x) = (v_0, v_1, \dots) \in \{0, 1\}^{\mathbb{N}}$ with $v_i \equiv T^i(x) \pmod{2}$. Bernstein and Lagarias [3, 2] showed that $x \mapsto v(x)$ is a bijection onto $\{0, 1\}^{\mathbb{N}}$, indeed a 2-adic isometry once $\{0, 1\}^{\mathbb{N}}$ is identified with $\mathbb{Z}_2$ through binary expansions; we write Φ for its inverse, so that $\Phi(v)$ is the unique 2-adic integer whose parity vector is v . Conjugating T by Φ turns it into the shift.

Rationality of $\Phi(v)$ corresponds to eventual periodicity of v in one direction only. If v is eventually periodic then $\Phi(v) \in \mathbb{Q}$; the converse is Lagarias’s Periodicity Conjecture [10, §2], which is open, and which asserts that no aperiodic parity vector has a rational Φ -value. Very little is known unconditionally in that direction. López and Stoll [11] studied Φ on Sturmian parity words and produced a 2-adic series and a generalised continued fraction for the values. They state in their abstract that it is unknown whether there exists any aperiodic v with

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an eventually periodic $\Phi(v)$, and that their computed examples suggest that Φ always maps Sturmian words to words of full complexity.

This paper proves a complexity-class result: no rational 2-adic integer has a parity word that is not eventually periodic and satisfies $\liminf_L p(L)/L < 1/\log_2(3/2) = 1.70951\dots$. The Sturmian statement that follows from it, $\Phi(s) \notin \mathbb{Q}$ for every Sturmian word, has two independent proofs here — a two-line counting corollary, and an elementary effective 2-adic argument. Let

$$\beta = \frac{\ln 2}{\ln 3} = \log_3 2 = 0.63092975\dots, \quad \alpha = 1/\beta = \log_2 3,$$

and let $1c_\beta$ be the Sturmian word of slope β and intercept 0 in the notation of [11], namely $1c_\beta(j) = \lceil (j+1)\beta \rceil - \lceil j\beta \rceil$ for $j \geq 0$. Its ones-density is β , the critical density for the $3x+1$ map: by (1) below, an orbit whose parity word has $k_\ell/\ell \rightarrow \beta$ satisfies $(3^{k_\ell}/2^\ell)^{1/\ell} \rightarrow 3^\beta/2 = 1$, so its multiplicative growth rate is exactly 1. This is the hardest case for the 2-adic argument: the real series attached to it diverges (Section 11).

Theorem 1.1. $\Phi(1c_\beta) \notin \mathbb{Q}$. *Equivalently, the 2-adic expansion of $\Phi(1c_\beta)$ is not eventually periodic.*

Theorem 1.2. $\Phi(s) \notin \mathbb{Q}$ for every Sturmian word s : every irrational slope $\gamma \in (0, 1)$, every intercept, and both mechanical words.

Theorem 1.1 is the case of the critical slope at intercept 0. Theorem 1.2 has two independent proofs in this paper: a short counting argument after Dubickas (Section 9, Corollary 9.4), and the elementary effective 2-adic argument of Section 10, which is what also yields Theorem 1.1 together with the sharper depth law and height floors that counting alone does not give.

The critical density is exactly where the existing methods stop. López and Stoll's irrationality results in [11] concern the *real* function $\Phi_{\mathbb{R}}$ and hold on $\mathbb{Q} \cap (\beta, 1]$, with a dual construction below β ; the restriction is forced, because the relevant real series converges if and only if $\beta < p/q \leq 1$, and their staircase diverges at $x = \beta$. Their later preprint [12] proves that Φ maps an aperiodic parity word whose *lower* ones-density is *strictly* greater than β to an aperiodic 2-adic integer, and its Theorem 1 shows that a divergent trajectory of a rational 2-adic integer would have lower ones-density exactly β . That Theorem 1 has two halves: the half above β is their own archimedean argument, while the half below β is quoted there [12, p. 6] from Monks and Yazinski [16, Theorem 2.7(b)], who proved that a rational 2-adic integer whose orbit is infinite has lower ones-density at least β . The word of Theorem 1.1 sits at the single density excluded by both halves, and the present argument is 2-adic rather than archimedean.

Method, counting. Section 9 gives a proof of Theorem 1.2 in two pages, after Dubickas [9, Theorem 5]: his complexity bound for positive-integer $3x+1$ trajectories transports unchanged to $\Phi(s)$ whenever $\Phi(s) \in \mathbb{Q}$, because T preserves a fixed odd denominator (Lemma 9.1), and a Sturmian word's complexity $p_s(L) = L+1$ then puts it below the resulting threshold $1/\log_2(3/2) = 1.70951\dots$ (Corollary 9.4). This gives the qualitative statement at every slope and intercept, with an explicit height bound, but nothing beyond it: no exact depth law, no height floor along a specific approximating sequence, no irrationality measure. The rest of the paper, ingredients (i)–(iii) below and Section 10, is an independent route to the same conclusion that also supplies those finer statements.

Method, Liouville route. Three ingredients, each elementary.

- (i) For a finite word w of length ℓ with k ones, the periodic word w^∞ has $\Phi(w^\infty) = c_w/(2^\ell - 3^k)$ with c_w an explicit positive integer (Proposition 4.1). When w is *mechanical*, $c_w \leq \ell \max(2^\ell, 3^k)$ (Lemma 4.2); for general w no such bound holds, at any constant (Remark 4.3).

- (ii) Let p_n/q_n be the convergents of β and w_n the mechanical word of slope p_n/q_n . Then $\Phi(w_n^\infty)$ agrees with $\Phi(1c_\beta)$ to 2-adic depth exactly $q_n - 1$ when $p_n/q_n < \beta$, and to depth exactly $q_n + q_{n+1} - 1$ when $p_n/q_n > \beta$ (Theorem 5.3). The asymmetry is the point: $1c_\beta$ is approached from one side, and the convergents on that side see one full continued-fraction level further.
- (iii) If $\Phi(1c_\beta) = u/v$ then the integer $M_n = u\delta_n - vc_n$ is nonzero with $v_2(M_n)$ equal to the depth in (ii), so $|M_n| \geq 2^{q_n + q_{n+1} - 1}$, while (i) gives $|M_n| \leq 4Hq_n2^{q_n}$ with $H = \max(|u|, v)$. Hence $q_{n+1} \leq 3 + \log_2 H + \log_2 q_n$, which fails for large n (Theorem 6.1).

No property of the continued fraction of β is used beyond $q_n \rightarrow \infty$: no irrationality measure for β , no bound on its partial quotients. The argument is effective, and yields an explicit height below which no rational can lie (Corollary 6.2).

Method, second argument. Ingredients (i) and (iii) are reused verbatim; ingredient (ii), the depth law, is replaced. A Sturmian word s begins in infinitely many squares WW with $|W| \rightarrow \infty$ (Theorem 10.2), and the primitive root W of such a square is a cyclic permutation of a standard word (Theorem 10.3), so W^∞ is balanced and (i) applies to it with the constant 3 of Remark 4.3. Since s and W^∞ share the prefix WW , (iii) runs with depth 2ℓ against height $\max(2^\ell, 3^k)$, and the surplus is $c(\gamma)\ell$ with $c(\gamma) = 2 - \max(1, \gamma \log_2 3) \geq 2 - \log_2 3$. No continued fraction is used at all, and no intercept is privileged.

Credit. The periodic approximants and their approximation depths are not new. For every irrational slope, [11, Theorem 1] gives a closed 2-adic series for $\Phi(1c_\beta)$ indexed by consecutive pairs of continued-fraction convergents, whose term exponents are $q_{j+1} + q_j - 1$; these are the depths of Theorem 5.3 at the convergents from above. A companion technical note of the author [7] derived the depths at both parities independently, and more sharply, in the coordinates of the accelerated $3x + 1$ map. What is new here is ingredient (iii), the Liouville comparison, and the elementary effective 2-adic proof at β ; the question of irrationality at this slope is stated as open in López–Stoll (2009) and treated as open in López–Stoll (2021), neither of which reaches it by any method.

Section 9 owes its content to Dubickas [9, Theorem 5]. His Theorem 5 concerns a sequence of positive integers $x_n = U(x_{n-1})$, $x_n \rightarrow \infty$, generated by a linear map U ; its parity sequence X has complexity $p_X(L) > 1.70951129L$ for all large L . The proof rests on two facts: that two equal length- m parity blocks, at times n and s , force $2^m \mid x_n - x_s$; and the growth bound $x_n < (3/2)^n(x_0 + 1)$. The scope of [9] is positive integers only, with nothing on rationals, negative integers, $\mathbb{Z}_2$, or the generalised $3x + q$ maps. The transport to every rational 2-adic integer (Lemma 9.1, Proposition 9.2) is not a theorem stated in [9]; it replaces $x_n \rightarrow \infty$ with the weaker fact that T preserves an odd denominator (Lemma 9.1), and replaces the divergence argument for distinctness of the x_n with aperiodicity of s (Proposition 9.2). This transport is ours; we have not found it stated elsewhere. The same mechanism — a repeated length- L factor forcing agreement of the underlying value modulo 2^L — appears independently in two 2026 sources: Stephan [18], whose Theorem 4.2 is this 2-adic rigidity statement for the single orbit of the minimal word of the rational base $3/2$, and whose §5.2 proves that a *rational* value of that orbit's limiting constant forces its complexity to exceed every linear bound, a stronger conclusion than Corollary 9.3 in that narrower setting; and Sharpe [17], an unrefereed GitHub repository whose `paper/parity_complexity.md` (first committed 2026-09-05) cites Dubickas explicitly and formalises, in Lean, the same rigidity-and-counting argument as a finite bound on positive-integer Collatz orbits. None of the three treats the general Sturmian statement of Theorem 1.2; that corollary of the counting bound, and the identification of the margin $2 - \log_2 3$ shared with Theorem 10.1 (Remark 9.5), are recorded here.

Away from β the credit is different, and Section 8 says so where the general theorem is proved. For irrational $\gamma < \beta$ the irrationality of $\Phi(1c_\gamma)$ is an immediate consequence of [16, Theorem 2.7(b)], drawn explicitly in [12, p. 6]; what Theorem 8.5 adds there is the effective height bound of Corollary 8.7, which a density argument does not give. For irrational $\gamma > \beta$ the statement is the first half of [12, Theorem 1], whose proof is archimedean and whose source is an unpublished preprint; Theorem 8.5 gives an elementary and effective 2-adic proof instead. At $\gamma = \beta$ it is the Liouville-route proof of Theorem 8.5 that is new; the statement itself already follows from Corollary 9.4, via the transport of Section 9.

The same division of credit governs Section 10. Its two combinatorial inputs are quoted, not proved here: that every Sturmian word begins in infinitely many squares is [1, Theorem 1] (see also [6]), and that the primitive root of such a square is a cyclic permutation of a standard word is [4, Proposition 3.2]. What that section contributes is the height bound for cyclic permutations of standard words (Lemma 10.4, one line from Remark 4.3), the Liouville comparison, and the observation that these two published facts are exactly what the comparison needs. For a word of ones-density $\gamma \neq \beta$ the irrationality conclusion was, again, already available from the density results quoted above, since a Sturmian word of slope γ has ones-density exactly γ . The statement itself, at every slope and intercept, already follows from Corollary 9.4; what is new here is the single elementary effective Liouville proof covering all Sturmian words uniformly, by initial squares rather than convergents, together with the height floor it supplies at every intercept.

Scope. Theorem 1.1 concerns one explicit word of critical density, together with its shifts and the two variants $c_\beta, 0c_\beta$. Section 8 runs the same three ingredients at an arbitrary irrational slope $\gamma \in (0, 1)$ and obtains $\Phi(1c_\gamma) \notin \mathbb{Q}$ with an explicit height bound (Theorem 8.5, Corollary 8.7); as recorded in the Credit paragraph and in Section 8, it is the Liouville-route proof there that is new at $\gamma = \beta$ specifically (the statement itself already following from Corollary 9.4), the contribution below β being the effectivity and the contribution above β being an elementary and effective proof in place of an unpublished archimedean one. Section 10 then removes the restriction to intercept 0 altogether, by the second argument, and obtains Theorem 1.2: Φ maps the whole Sturmian subshift X_γ — an uncountable set, against the countably many words per slope reached by the first argument — into the irrationals (Corollary 10.5). It says nothing about divergent orbits of positive integers, nothing about the Periodicity Conjecture in general — which quantifies over all aperiodic parity vectors, an uncountable family — and nothing about the Collatz conjecture. The paper is self-contained: every object is defined here, and no machinery from [7, 8] is needed to read it.

Section 9 marks the frontier of the counting route precisely: Corollary 9.3 reaches every word with $\liminf_L p_s(L)/L < 1/\log_2(3/2) = 1.70951\dots$, and no further. Rote words and binary codings of rotations with two intervals have $p_s(L) = 2L$, above the threshold, and remain open for this method (Section 12, problem (4)); the repetition method of Sections 6–10 is what carries the load for Sturmian words specifically, and gives besides the exact depth law, the height floors of Corollary 6.2 and Corollary 8.7, and the setting for the irrationality-measure question of Section 12, problem (2), none of which the counting bound addresses.

2. THE CONJUGACY MAP

Throughout, v_2 is the 2-adic valuation on $\mathbb{Z}_2$ with $v_2(0) = \infty$, and for $v, v' \in \{0, 1\}^\mathbb{N}$ we write $\text{lcp}(v, v')$ for the length of their longest common prefix. For $v \in \{0, 1\}^\mathbb{N}$ and $m \geq 0$ put

$$k_m(v) = \#\{i < m : v_i = 1\},$$

the number of ones among $v_0, \dots, v_{m-1}$; thus $k_0(v) = 0$.

The one-step identity $T(x) = (3^{v_0}x + v_0)/2$, valid for both parities, gives by induction the standard formula

$$T^n(x) = \frac{3^{k_n}x + c_n}{2^n}, \quad c_n = c_n(v) = \sum_{i < n, v_i=1} 3^{k_n - k_{i+1}} 2^i, \quad (1)$$

where $v = v(x)$ and $k_m = k_m(v)$. Indeed $c_0 = 0$ and $c_{n+1} = 3^{v_n}c_n + v_n 2^n$.

Proposition 2.1. *Let $v \in \{0, 1\}^{\mathbb{N}}$ contain infinitely many ones. Then the series below converges in $\mathbb{Z}_2$ and*

$$\Phi(v) = - \sum_{i: v_i=1} 3^{-k_{i+1}(v)} 2^i.$$

Proof. Each $3^{-k_{i+1}}$ is a 2-adic unit and $v_2(3^{-k_{i+1}} 2^i) = i$, so the terms tend to 0 and the series converges. Let $x = \Phi(v)$. Rearranging (1) gives $3^{k_n}x = 2^n T^n(x) - c_n$, whence

$$x = \frac{2^n T^n(x)}{3^{k_n}} - \frac{c_n}{3^{k_n}} = \frac{2^n T^n(x)}{3^{k_n}} - \sum_{i < n, v_i=1} 3^{-k_{i+1}} 2^i.$$

Since $T^n(x) \in \mathbb{Z}_2$ and 3^{-k_n} is a unit, the first term has valuation at least n , so it tends to 0; letting $n \rightarrow \infty$ gives the claim. $\square$

Proposition 2.2. *For all $v, v' \in \{0, 1\}^{\mathbb{N}}$,*

$$v_2(\Phi(v) - \Phi(v')) = \text{lcp}(v, v').$$

Proof. Fix $L \geq 1$. By (1) with $n = L$ we have $3^{k_L}x \equiv -c_L \pmod{2^L}$, so $x \equiv -c_L 3^{-k_L} \pmod{2^L}$; and k_L, c_L depend only on $v_0, \dots, v_{L-1}$. Hence $x \pmod{2^L}$ is determined by the first L letters of $v(x)$. Conversely $v_0, \dots, v_{L-1}$ are determined by $x \pmod{2^L}$, by induction on L : $v_0 \equiv x \pmod{2}$, and $T(x) \pmod{2^{L-1}}$ is determined by $x \pmod{2^L}$. So $v \mapsto \Phi(v) \pmod{2^L}$ is a bijection from $\{0, 1\}^L$ onto $\mathbb{Z}/2^L\mathbb{Z}$. Consequently $\Phi(v) \equiv \Phi(v') \pmod{2^L}$ if and only if v and v' agree in their first L letters, which is the assertion. $\square$

Proposition 2.3. *Let σ denote the shift on $\{0, 1\}^{\mathbb{N}}$. Then $\Phi(\sigma v) = T(\Phi(v))$, and $\Phi(v)$ is odd if and only if $v_0 = 1$.*

Proof. The parity vector of $T(x)$ is $\sigma v(x)$ by definition, and $v_0(x) \equiv x \pmod{2}$. $\square$

3. THE CRITICAL STURMIAN WORD

Let $\beta = \log_3 2 \in (0, 1)$ and $\alpha = 1/\beta = \log_2 3$, both irrational. In the notation of [11],

$$1c_\beta(j) = \lceil (j+1)\beta \rceil - \lceil j\beta \rceil, \quad 0c_\beta(j) = \lfloor (j+1)\beta \rfloor - \lfloor j\beta \rfloor \quad (j \geq 0),$$

and c_β denotes the characteristic word $c_\beta(j) = \lfloor (j+1)\beta \rfloor - \lfloor j\beta \rfloor$ for $j \geq 1$. Since $\beta \in (0, 1)$ is irrational, $1c_\beta = 1c_\beta$ and $0c_\beta = 0c_\beta$.

Lemma 3.1. *For $n \geq 0$ we have $k_n(1c_\beta) = \lceil n\beta \rceil$, and for $j \geq 0$ the $(j+1)$ -st one of $1c_\beta$ occupies position $\lfloor j\alpha \rfloor$.*

Proof. The first claim is a telescoping sum. For the second, position i carries the $(j+1)$ -st one exactly when $k_i(1c_\beta) = j$ and $k_{i+1}(1c_\beta) = j+1$, that is, $\lceil i\beta \rceil \leq j$ and $\lceil (i+1)\beta \rceil \geq j+1$. For $j \geq 1$ the first reads $i\beta \leq j$, i.e. $i \leq j\alpha$, i.e. $i \leq \lfloor j\alpha \rfloor$ since $j\alpha \notin \mathbb{Z}$; the second reads $(i+1)\beta > j$, i.e. $i+1 > j\alpha$, i.e. $i \geq \lfloor j\alpha \rfloor$. Hence $i = \lfloor j\alpha \rfloor$. For $j = 0$ both conditions force $i = 0 = \lfloor 0 \cdot \alpha \rfloor$. $\square$

Proposition 3.2. *Let $\Xi_\alpha = \sum_{j \geq 0} 3^{-(j+1)} 2^{\lfloor j\alpha \rfloor} \in \mathbb{Z}_2$. Then $\Phi(1c_\beta) = -\Xi_\alpha$. In particular $\Phi(1c_\beta)$ is a 2-adic unit.*

Proof. By Lemma 3.1 the ones of $1c_\beta$ are at the positions $\lfloor j\alpha \rfloor$, $j \geq 0$, and the one at position $\lfloor j\alpha \rfloor$ is the $(j+1)$ -st, so $k_{i+1} = j+1$ there. Proposition 2.1 gives the formula. The term $j=0$ has valuation 0 and all others have positive valuation, so $v_2(\Xi_\alpha) = 0$. $\square$

The constant Ξ_α is the 2-adic Sturmian carry constant of [7]; Proposition 3.2 is the dictionary between the two normalisations, and we shall not need it again. All statements below are about Φ .

4. PERIODIC VALUES AND THEIR HEIGHTS

For a finite word $w = w_0 \cdots w_{\ell-1} \in \{0, 1\}^\ell$ with $k \geq 1$ ones we write w^∞ for its infinite periodic repetition and put

$$c_w = \sum_{i < \ell, w_i=1} 3^{k-k_{i+1}(w)} 2^i \in \mathbb{Z}_{>0}. \quad (2)$$

Proposition 4.1. *With the notation above, $2^\ell \neq 3^k$ and*

$$\Phi(w^\infty) = \frac{c_w}{2^\ell - 3^k}.$$

Proof. $2^\ell = 3^k$ is impossible for $k \geq 1$ by unique factorisation. Write $i = t\ell + r$ with $0 \leq r < \ell$; periodicity gives $k_{i+1}(w^\infty) = tk + k_{r+1}(w)$. Grouping the sum of Proposition 2.1 by periods,

$$\Phi(w^\infty) = - \sum_{t \geq 0} \sum_{r < \ell, w_r=1} 3^{-(tk+k_{r+1}(w))} 2^{t\ell+r} = - \left(\sum_{r < \ell, w_r=1} 3^{-k_{r+1}(w)} 2^r \right) \sum_{t \geq 0} (3^{-k} 2^\ell)^t.$$

The geometric series converges in $\mathbb{Z}_2$ because $v_2(3^{-k} 2^\ell) = \ell \geq 1$, with sum $(1 - 3^{-k} 2^\ell)^{-1} = 3^k / (3^k - 2^\ell)$. Since $c_w = 3^k \sum_{r < \ell, w_r=1} 3^{-k_{r+1}(w)} 2^r$, multiplying gives $\Phi(w^\infty) = -c_w / (3^k - 2^\ell)$, which is the stated value. $\square$

For coprime $1 \leq p \leq q$ let $w_{p,q}$ be the *mechanical word of slope p/q* : the word of length q whose ones occupy the positions $\lfloor jq/p \rfloor$ for $0 \leq j < p$. It has exactly p ones.

Lemma 4.2. *Let $w = w_{p,q}$. Then $0 < c_w \leq q \cdot \max(2^q, 3^p)$.*

Proof. Positivity is clear from (2). For $0 \leq i < q$,

$$k_{i+1}(w) = \#\{0 \leq j < p : \lfloor jq/p \rfloor \leq i\} = \#\{0 \leq j < p : j < (i+1)p/q\} \geq (i+1)p/q,$$

the last step because the count is $\min(p, \lceil (i+1)p/q \rceil)$ and $(i+1)p/q \leq p$. Hence $p - k_{i+1}(w) \leq p(q-1-i)/q$ and, with $\rho = 3^{p/q}$,

$$3^{p-k_{i+1}(w)} 2^i \leq \rho^{q-1-i} 2^i \leq \max(\rho^{q-1}, 2^{q-1}) \leq \max(3^p, 2^q),$$

where the middle inequality holds because $i \mapsto \rho^{q-1-i} 2^i$ is monotone, so its maximum over $0 \leq i \leq q-1$ is attained at an endpoint. The sum (2) has at most q terms. $\square$

Remark 4.3. The hypothesis that w be mechanical cannot be dropped, and not merely the constant is at stake. Take $w = 0^a 1^a$, so $\ell = 2a$ and $k = a$. Its ones are at $i = a, \dots, 2a-1$ with $k_{i+1}(w) = i - a + 1$, so by (2)

$$c_w = \sum_{t=0}^{a-1} 3^{a-1-t} 2^{a+t} = 2^a \sum_{t=0}^{a-1} 3^{a-1-t} 2^t = 2^a (3^a - 2^a).$$

Here $\max(2^\ell, 3^k) = 4^a$, so

$$\frac{c_w}{\ell \cdot \max(2^\ell, 3^k)} = \frac{3^a - 2^a}{2a \cdot 2^a} = \frac{(3/2)^a - 1}{2a} \xrightarrow{a \rightarrow \infty} \infty.$$

Thus no bound $c_w \leq C\ell \max(2^\ell, 3^k)$ holds for arbitrary w , for any constant C . Numerically the ratio is 2.83 at $a = 10$, 14.56 at $a = 15$ and 41.03 at $a = 18$. For an arbitrary *balanced* w the proof of Lemma 4.2 goes through with the discrepancy bound $|k_{i+1}(w) - (i+1)k/\ell| < 1$ in place of the exact count, at the cost of a factor 3.

5. THE APPROXIMATION DEPTH

Convention 5.1. p_n/q_n denotes the n -th convergent of the continued fraction

$$\beta = [0; 1, 1, 1, 2, 2, 3, 1, 5, 2, 23, \dots],$$

indexed from $p_0/q_0 = 0/1$, so that $p_1/q_1 = 1/1$ and $p_2/q_2 = 1/2$. Thus p_n counts ones and q_n counts letters. We abbreviate $w_n = w_{p_n, q_n}$ and $\alpha_n = q_n/p_n$, and write $D_n = |q_n\beta - p_n|$, so that the classical estimate $D_n < 1/q_{n+1}$ holds.

The first convergents are

n	1	2	3	4	5	6	7
p_n/q_n	1/1	1/2	2/3	5/8	12/19	41/65	53/84

n	8	9	10	11	12	13	14
p_n/q_n	$\frac{306}{485}$	$\frac{665}{1054}$	$\frac{15601}{24727}$	$\frac{31867}{50508}$	$\frac{79335}{125743}$	$\frac{111202}{176251}$	$\frac{190537}{301994}$

Because the convergents alternate about β and $p_1/q_1 = 1 > \beta$, we have $p_n/q_n > \beta$ exactly when n is odd. Equivalently, comparing exactly, $3^{p_n} > 2^{q_n}$ if and only if n is odd. Also $q_{n+1} > q_n$ for $n \geq 1$.

Lemma 5.2 (floor comparison). *Let $n \geq 3$ and $j_n^* = \min\{j \geq 1 : \lfloor j\alpha \rfloor \neq \lfloor j\alpha_n \rfloor\}$.*

- (1) *If n is odd, then $j_n^* = p_n + p_{n+1}$, and at that index $\lfloor j_n^*\alpha \rfloor = q_n + q_{n+1}$ while $\lfloor j_n^*\alpha_n \rfloor = q_n + q_{n+1} - 1$.*
- (2) *If n is even, then $j_n^* = p_n$, and at that index $\lfloor j_n^*\alpha \rfloor = q_n - 1$ while $\lfloor j_n^*\alpha_n \rfloor = q_n$.*

Proof. Write $\alpha_n - \alpha = \eta_n$. Since $\alpha_n = q_n/p_n$ and $\alpha = 1/\beta$,

$$\eta_n = \frac{q_n}{p_n} - \frac{1}{\beta} = \frac{q_n\beta - p_n}{p_n\beta},$$

so $\eta_n < 0$ when n is odd and $\eta_n > 0$ when n is even, and in both cases $|\eta_n| = D_n/(p_n\beta)$.

Set $m_j = jq_n \bmod p_n$, so that $\{j\alpha_n\} = m_j/p_n$; since $\gcd(p_n, q_n) = 1$ we have $m_j \geq 1$ for $1 \leq j < p_n$, and $m_{p_n} = 0$.

(2), n even. Over the range $j \leq p_n$ used here the two floor sequences differ by at most one, since $j|\eta_n| \leq p_n|\eta_n| = \alpha D_n < \alpha/q_{n+1} < 1$; the equivalence below is therefore valid. Here $\eta_n > 0$, so a mismatch at j means $\lfloor j\alpha_n \rfloor > \lfloor j\alpha \rfloor$, which requires $\{j\alpha_n\} < j\eta_n$, i.e. $m_j < jp_n\eta_n = jD_n/\beta = j\alpha D_n$. For $1 \leq j < p_n$ this fails, because $m_j \geq 1$ while $j\alpha D_n < p_n\alpha D_n < p_n\alpha/q_{n+1} \leq 1$, using $D_n < 1/q_{n+1}$ and $p_n\alpha < q_n < q_{n+1}$. At $j = p_n$ it holds, since $m_{p_n} = 0$. So $j_n^* = p_n$. The depths: $p_n\alpha_n = q_n$ exactly, so $\lfloor p_n\alpha_n \rfloor = q_n$; and $p_n\alpha = q_n - \alpha D_n$ with $0 < \alpha D_n < \alpha/q_{n+1} < 1$, so $\lfloor p_n\alpha \rfloor = q_n - 1$.

(1), n odd. Here $\eta_n < 0$, so a mismatch at j means $\lfloor j\alpha_n \rfloor < \lfloor j\alpha \rfloor$, which happens exactly when $\{j\alpha_n\} + j|\eta_n| \geq 1$, i.e.

$$m_j + j\alpha D_n \geq p_n. \tag{3}$$

We use two exact facts about consecutive convergents. First, $p_{n+1}q_n - p_nq_{n+1} = (-1)^n = -1$, so $p_{n+1}q_n \equiv -1 \pmod{p_n}$ and

$$m_{p_{n+1}} = p_n - 1, \quad p_{n+1}\alpha_n = q_{n+1} - \frac{1}{p_n}. \quad (4)$$

Second, writing $p_m = \beta q_m + (-1)^{m+1}D_m$ and using the classical identity $q_{n+1}D_n + q_nD_{n+1} = 1$,

$$(p_n + p_{n+1})D_n = \beta(q_n + q_{n+1})D_n + D_n(D_n - D_{n+1}) = \beta + (D_n - D_{n+1})(\beta q_n + D_n),$$

so, multiplying by $\alpha = 1/\beta$ and using $p_n = \beta q_n + D_n$,

$$(p_n + p_{n+1})\alpha D_n = 1 + p_n\alpha(D_n - D_{n+1}). \quad (5)$$

Note $0 < p_n\alpha(D_n - D_{n+1}) < (q_n + 1)D_n \leq (q_n + 1)/q_{n+1} \leq 1$, because $p_n\alpha = q_n + \alpha D_n < q_n + 1$ and $q_{n+1} \geq q_n + 1$; so the left side of (5) lies in $(1, 2)$. Subtracting $p_n\alpha D_n$ from (5) gives moreover

$$p_{n+1}\alpha D_n = 1 - p_n\alpha D_{n+1} < 1. \quad (6)$$

Finally, over the range of indices used below the two floor sequences differ by at most one: for $j \leq p_n + p_{n+1}$,

$$j|\eta_n| = \frac{j\alpha D_n}{p_n} \leq \frac{(p_n + p_{n+1})\alpha D_n}{p_n} < \frac{2}{p_n} \leq 1,$$

since $p_n \geq 2$ for $n \geq 3$. This is what makes (3) an equivalence.

Now no $j \leq p_{n+1}$ satisfies (3): since $m_j \leq p_n - 1$, it would force $j\alpha D_n \geq 1$, whereas $j\alpha D_n \leq p_{n+1}\alpha D_n < 1$ by (6). Next, for $j = p_{n+1} + i$ with $1 \leq i \leq p_n - 1$ we have $m_j = (iq_n \bmod p_n) - 1 \leq p_n - 2$ by (4), since $iq_n \not\equiv 0 \pmod{p_n}$; and $j\alpha D_n < (p_n + p_{n+1})\alpha D_n < 2$, so $m_j + j\alpha D_n < (p_n - 2) + 2 = p_n$ and (3) again fails. At $j^* = p_n + p_{n+1}$, however, $m_{j^*} = m_{p_{n+1}} = p_n - 1$ and, by (5),

$$m_{j^*} + j^*\alpha D_n = (p_n - 1) + 1 + p_n\alpha(D_n - D_{n+1}) = p_n + p_n\alpha(D_n - D_{n+1}) > p_n,$$

since $D_n > D_{n+1}$. Hence $j_n^* = p_n + p_{n+1}$.

The two depths are now exact. By (4), $j^*\alpha_n = q_n + p_{n+1}\alpha_n = q_n + q_{n+1} - 1/p_n$, and $p_n \geq 2$, so $\lfloor j^*\alpha_n \rfloor = q_n + q_{n+1} - 1$. Moreover $j^*\alpha = j^*\alpha_n + j^*\alpha D_n/p_n$, and by (5) $j^*\alpha D_n/p_n = 1/p_n + \alpha(D_n - D_{n+1})$, whence

$$j^*\alpha = q_n + q_{n+1} + \alpha(D_n - D_{n+1}), \quad 0 < \alpha(D_n - D_{n+1}) < \alpha D_n < \alpha/q_{n+1} < 1,$$

so $\lfloor j^*\alpha \rfloor = q_n + q_{n+1}$. □

Theorem 5.3 (depth law). *For every $n \geq 3$,*

$$v_2(\Phi(1c_\beta) - \Phi(w_n^\infty)) = \text{lcp}(1c_\beta, w_n^\infty) = \begin{cases} q_n - 1, & n \text{ even}, \\ q_n + q_{n+1} - 1, & n \text{ odd}. \end{cases}$$

Proof. The first equality is Proposition 2.2. For the second, both words are determined by their sets of one-positions: by Lemma 3.1 the ones of $1c_\beta$ are at $\lfloor j\alpha \rfloor$, $j \geq 0$, and by periodicity the ones of w_n^∞ are at $\lfloor j\alpha_n \rfloor$, $j \geq 0$ — indeed $\lfloor (j + p_n)\alpha_n \rfloor = \lfloor j\alpha_n \rfloor + q_n$. Both position sequences are strictly increasing, and they agree for $j < j_n^*$ by definition of j_n^* . Hence the two words agree at every position below $\min(\lfloor j_n^*\alpha \rfloor, \lfloor j_n^*\alpha_n \rfloor)$ and differ at that position, one of them carrying a one there and the other a zero. So the longest common prefix has length $\min(\lfloor j_n^*\alpha \rfloor, \lfloor j_n^*\alpha_n \rfloor)$, which Lemma 5.2 evaluates as $q_n + q_{n+1} - 1$ for odd n and $q_n - 1$ for even n . □

Remark 5.4. The two cases differ by a whole continued-fraction level, and only the odd case is of any use below: for even n the depth $q_n - 1$ is smaller than the archimedean size of the approximant, and the comparison of Section 6 is vacuous there. The reason for the asymmetry is that $1c_\beta$ is the limit of the mechanical words from one side, so a periodic extension taken from that side continues to agree with it for one level longer.

6. IRRATIONALITY

Theorem 6.1. $\Phi(1c_\beta) \notin \mathbb{Q}$. More precisely, suppose $\Phi(1c_\beta) = u/v$ with $u \in \mathbb{Z}$, $v \geq 1$, $\gcd(u, v) = 1$, and put $H = \max(|u|, v)$. Then every odd $n \geq 3$ satisfies

$$q_{n+1} \leq 3 + \log_2 H + \log_2 q_n. \quad (7)$$

Since $q_{n+1} > q_n$ and $q_n \rightarrow \infty$ along odd n , this is impossible; a contradiction has appeared by the first odd $n \geq 3$ with $q_n \geq 2 \log_2 H + 20$.

Proof. Assume $\Phi(1c_\beta) = u/v$ as stated. As $\Phi(1c_\beta) \in \mathbb{Z}_2$, the denominator v is odd. Fix an odd $n \geq 3$ and write

$$\delta_n = 2^{q_n} - 3^{p_n}, \quad c_n = c_{w_n}, \quad \Phi(w_n^\infty) = \frac{c_n}{\delta_n}$$

by Proposition 4.1. Since n is odd, $3^{p_n} > 2^{q_n}$, so $\delta_n < 0$ and $|\delta_n| = 3^{p_n} - 2^{q_n} < 3^{p_n} = \max(2^{q_n}, 3^{p_n})$. Put

$$M_n = u\delta_n - v c_n \in \mathbb{Z}, \quad \text{so} \quad \Phi(1c_\beta) - \Phi(w_n^\infty) = \frac{M_n}{v\delta_n}.$$

M_n is a nonzero integer of known valuation. Both v and δ_n are odd, so $v_2(M_n) = v_2(\Phi(1c_\beta) - \Phi(w_n^\infty))$, which Theorem 5.3 evaluates as $q_n + q_{n+1} - 1$. In particular the valuation is finite, so $M_n \neq 0$, and therefore

$$|M_n| \geq 2^{q_n + q_{n+1} - 1}. \quad (8)$$

Archimedean size. By Lemma 4.2, $c_n \leq q_n 3^{p_n}$. Hence

$$|M_n| \leq |u| |\delta_n| + v c_n \leq H \cdot 3^{p_n} + H \cdot q_n \cdot 3^{p_n} = H(1 + q_n)3^{p_n} \leq 2Hq_n 3^{p_n}.$$

Now $3^{p_n} = 2^{p_n \alpha}$ and $p_n \alpha - q_n = \alpha(p_n - q_n \beta) = \alpha D_n$, with $0 < \alpha D_n < \alpha/q_{n+1} \leq \alpha/8 < 0.2$ because $n \geq 3$ forces $q_{n+1} \geq 8$. So $3^{p_n} < 2^{q_n + 0.2} < 1.15 \cdot 2^{q_n}$ and

$$|M_n| < 2.3 H q_n 2^{q_n} \leq 4Hq_n 2^{q_n}. \quad (9)$$

Comparison. Combining (8) and (9), $2^{q_n + q_{n+1} - 1} < 4Hq_n 2^{q_n}$, that is $2^{q_{n+1} - 1} < 4Hq_n$, and taking logarithms gives (7).

Finally, (7) together with $q_{n+1} > q_n$ gives

$$q_n - \log_2 q_n < 3 + \log_2 H. \quad (10)$$

Suppose $q_n \geq 2 \log_2 H + 20$ and put $Q = q_n$. Then $\log_2 H \leq (Q - 20)/2$, so the right-hand side of (10) is at most $3 + (Q - 20)/2 = Q/2 - 7$, while the left-hand side exceeds $Q/2 - 7$ because $Q/2 + 7 > \log_2 Q$ for every $Q \geq 1$. This contradicts (10). Since $t \mapsto t - \log_2 t$ is increasing for $t \geq 2$, every odd n with $q_n \geq 2 \log_2 H + 20$ is impossible; as there are infinitely many odd n and $q_n \rightarrow \infty$, such an n exists. $\square$

Corollary 6.2. $\Phi(1c_\beta)$ is not equal to any u/v in lowest terms with $\max(|u|, v) \leq 2^{301973}$.

Proof. Take $n = 13$, so $p_n/q_n = 111202/176251$ and $q_{n+1} = 301994$; the depth of Theorem 5.3 is $q_n + q_{n+1} - 1 = 478244$, a value independently confirmed by the exact computation of Appendix A. By (8) and (9), $2^{478244} \leq |M_{13}| < 4H \cdot 176251 \cdot 2^{176251}$, so $H > 2^{478244 - 176251} / 705004 = 2^{301993} / 705004 > 2^{301973}$, since $705004 < 2^{20}$. $\square$

Remark 6.3. The argument uses no property of the continued fraction of β beyond $q_n \rightarrow \infty$. In particular it needs no irrationality measure for β and no bound on its partial quotients, and it is effective: by Theorem 6.1 the contradiction appears at continued-fraction depth $O(\log \log H)$.

7. CONSEQUENCES

Corollary 7.1. $\Phi(c_\beta)$ and $\Phi(0c_\beta)$ are irrational.

Proof. By Proposition 2.3, $\Phi(c_\beta) = T(\Phi(1c_\beta))$ and $\Phi(c_\beta) = T(\Phi(0c_\beta))$, since $\sigma(1c_\beta) = \sigma(0c_\beta) = c_\beta$. As $1c_\beta$ begins with 1, the value $\Phi(1c_\beta)$ is odd and $\Phi(c_\beta) = (3\Phi(1c_\beta) + 1)/2$; as $0c_\beta$ begins with 0, the value $\Phi(0c_\beta)$ is even and $\Phi(c_\beta) = \Phi(0c_\beta)/2$. Both relations are $\mathbb{Q}$ -affine with rational coefficients and invertible, so the three values are rational or irrational together. Theorem 6.1 decides which. $\square$

Corollary 7.2. $\Phi(\sigma^k 1c_\beta)$ is irrational for every $k \geq 0$.

Proof. $\Phi(\sigma^k 1c_\beta) = T^k(\Phi(1c_\beta))$ by Proposition 2.3. It therefore suffices that $T(x) \in \mathbb{Q}$ if and only if $x \in \mathbb{Q}$. If $x \in \mathbb{Q} \cap \mathbb{Z}_2$ then $T(x) \in \mathbb{Q} \cap \mathbb{Z}_2$. Conversely, T is two-to-one on $\mathbb{Z}_2$: the preimages of y are $2y$, which is even, and $(2y - 1)/3$, which lies in $\mathbb{Z}_2$ because 3 is a unit and is odd; the parity of x selects the branch. Both expressions are rational when y is. $\square$

Corollary 7.3. Let $q \geq 1$ be odd and let T_q be the generalised map $T_q(n) = n/2$ for n even and $T_q(n) = (3n + q)/2$ for n odd. There is no $n_0 \in \mathbb{Z}$ whose T_q -orbit has parity word $1c_\beta$, nor any whose parity word is a shift $\sigma^k 1c_\beta$.

Proof. Suppose $n_0 \in \mathbb{Z}$ has T_q -orbit $(n_j)_{j \geq 0}$ with parity word w , that is, n_j is odd exactly when $w_j = 1$. Put $x_j = n_j/q \in \mathbb{Q}$. Then x_j is odd in $\mathbb{Z}_2$ exactly when n_j is odd, because q is odd, and

$$x_{j+1} = \frac{n_{j+1}}{q} = \frac{3n_j + w_j q}{2q} = \frac{3x_j + w_j}{2},$$

so $x_{j+1} = T(x_j)$ and the parity vector of x_0 is w . Hence $\Phi(w) = x_0 \in \mathbb{Q}$. Conversely, if $\Phi(w) = u/v$ in lowest terms with v odd, then $n_j = vT^j(\Phi(w))$ is an integer T_v -orbit with parity word w . The corollary is therefore equivalent to the irrationality of $\Phi(w)$, which is Theorem 6.1 for $w = 1c_\beta$ and Corollary 7.2 for its shifts. $\square$

8. ALL IRRATIONAL SLOPES

Theorem 6.1 concerns one slope. The argument in fact needs nothing about β at all, and this section proves the corresponding statement for every irrational slope. We begin by saying precisely what is, and what is not, new in that generality.

What is known. Write $\gamma \in (0, 1)$ for an irrational slope and recall that $k_n(1c_\gamma) = \lceil n\gamma \rceil$, so that $1c_\gamma$ has ones-density exactly γ . Monks and Yazinski [16, Theorem 2.7(b)] proved that if $x \in \mathbb{Q} \cap \mathbb{Z}_2$ has a T -orbit which is not eventually cyclic — equivalently, whose orbit set is infinite — then

$$\liminf_{n \rightarrow \infty} \frac{k_n(v(x))}{n} \geq \beta.$$

Since a word is eventually periodic if and only if its Φ -value has an eventually cyclic orbit, this gives at once: *if v is aperiodic and $\liminf k_n(v)/n < \beta$, then $\Phi(v) \notin \mathbb{Q}$.* In particular $\Phi(1c_\gamma) \notin \mathbb{Q}$ for every irrational $\gamma < \beta$. The deduction is made explicitly by López and Stoll [12, p. 6]. Above β the corresponding statement — if v is aperiodic and $\liminf k_n(v)/n > \beta$ then $\Phi(v) \notin \mathbb{Q}$ — is the first half of [12, Theorem 1], proved there by an archimedean argument about accumulation points of the orbit of a real-valued $\Phi_{\mathbb{R}}(v)$; that preprint is unpublished. At $\gamma = \beta$ both statements are vacuous, which is why Theorem 6.1 needed a different proof.

What this section adds. Theorem 8.5 below proves $\Phi(1c_\gamma) \notin \mathbb{Q}$ for *every* irrational $\gamma \in (0, 1)$ by the single 2-adic argument of Sections 5–6, with $n_0 = 3$ uniformly in γ and with explicit constants. At $\gamma = \beta$ it is Theorem 6.1. For $\gamma > \beta$ it is an elementary and effective substitute

for the argument of [12]. For $\gamma < \beta$ the qualitative statement is [16], and what is new is the effective height bound of Corollary 8.7, which that density argument does not provide. The three regimes are proved here by one inequality, with no case distinction.

Notation.

Convention 8.1. Let $\gamma \in (0, 1)$ be irrational and $\alpha_\gamma = 1/\gamma > 1$. Let p_n/q_n be the convergents of $\gamma = [0; a_1, a_2, \dots]$, indexed from $p_0/q_0 = 0/1$, so $p_1/q_1 = 1/a_1$, and put $D_n = |q_n\gamma - p_n|$, $w_n = w_{p_n, q_n}$, $\alpha_n = q_n/p_n$. We use throughout the classical facts $D_n < 1/q_{n+1}$, $q_{n+1}D_n + q_nD_{n+1} = 1$, $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$, $D_{n+1} < D_n$, $q_{n+1} > q_n$ for $n \geq 1$, and $p_n = \gamma q_n + (-1)^{n+1}D_n$; in particular $p_n/q_n > \gamma$ if and only if n is odd. For $\gamma = \beta$ this is Convention 5.1 and $\alpha_\beta = \alpha = \log_2 3$. Finally set

$$\theta = \gamma \log_2 3, \quad \kappa = \max(1, \theta) - 1, \quad c(\gamma) = 2 - \max(1, \theta) = 1 - \kappa.$$

Note $\theta = 1$ exactly when $\gamma = \beta$, that $c(\gamma) = 1$ for $\gamma \leq \beta$, and that c decreases on $[\beta, 1)$ to the limit $2 - \log_2 3 = 0.41503\dots > 0$, which is not attained.

The depth law at a general slope. Lemma 3.1 and Lemma 4.2 are already stated for an arbitrary slope and an arbitrary mechanical word respectively, and their proofs use no property of β ; we use them below without further comment. The only input of Lemma 5.2 that is not a continued-fraction identity is the bound $\alpha D_n < 1$, and it is free:

Lemma 8.2. *For every irrational $\gamma \in (0, 1)$ and every $n \geq 1$ we have $q_{n+1} > \alpha_\gamma$, and hence $\alpha_\gamma D_n < \alpha_\gamma/q_{n+1} < 1$.*

Proof. From $\gamma = [0; a_1, a_2, \dots]$ we get $\alpha_\gamma = 1/\gamma = [a_1; a_2, a_3, \dots]$, so $a_1 < \alpha_\gamma < a_1 + 1$. Now $q_1 = a_1$ and $q_2 = a_2q_1 + q_0 = a_2a_1 + 1 \geq a_1 + 1 > \alpha_\gamma$. Since $(q_m)_{m \geq 1}$ is strictly increasing, $q_{n+1} \geq q_2 > \alpha_\gamma$ for all $n \geq 1$. The second assertion is $D_n < 1/q_{n+1}$. $\square$

Theorem 8.3 (depth law, every irrational slope). *Let $\gamma \in (0, 1)$ be irrational. For every $n \geq 3$,*

$$v_2(\Phi(1c_\gamma) - \Phi(w_n^\infty)) = \text{lcp}(1c_\gamma, w_n^\infty) = \begin{cases} q_n - 1, & n \text{ even}, \\ q_n + q_{n+1} - 1, & n \text{ odd}, \end{cases}$$

and the first index at which the two one-position sequences disagree is $j_n^ = p_n$ for n even and $j_n^* = p_n + p_{n+1}$ for n odd.*

Proof. The proofs of Lemma 5.2 and Theorem 5.3 apply verbatim with β replaced by γ and α by α_γ . Every step there is either one of the continued-fraction identities of Convention 8.1 or an instance of one of the following three facts, which are the only places where a property of the slope was used.

(i) *The side.* $\eta_n = \alpha_n - \alpha_\gamma = (q_n\gamma - p_n)/(p_n\gamma)$ is negative for n odd and positive for n even, with $|\eta_n| = \alpha_\gamma D_n/p_n$. This is the alternation of the convergents, valid for every irrational $\gamma \in (0, 1)$ with the indexing of Convention 8.1.

(ii) $\alpha_\gamma D_n < 1$. This is Lemma 8.2, for all $n \geq 1$. It is what is needed at each of the four places where the original proof wrote $\alpha D_n < \alpha/q_{n+1} < 1$: the “differ by at most one” bounds in both cases, the evaluation $\lfloor p_n \alpha_\gamma \rfloor = q_n - 1$ for n even, the estimate $p_n \alpha_\gamma = q_n + \alpha_\gamma D_n < q_n + 1$ giving $(p_n + p_{n+1})\alpha_\gamma D_n < 2$ for n odd, and the evaluation $\lfloor j^* \alpha_\gamma \rfloor = q_n + q_{n+1}$.

(iii) $p_n \geq 2$ for $n \geq 3$. Indeed $p_1 = 1$, $p_2 = a_2$, $p_3 = a_3a_2 + 1 \geq 2$, and $p_{n+1} = a_{n+1}p_n + p_{n-1} \geq p_n$ for $n \geq 2$. This is what makes $j|\eta_n| < 2/p_n \leq 1$ for $j \leq p_n + p_{n+1}$, and what gives $\lfloor j^* \alpha_n \rfloor = \lfloor q_n + q_{n+1} - 1/p_n \rfloor = q_n + q_{n+1} - 1$.

Finally the two one-position sequences are strictly increasing because $\alpha_\gamma > 1$ and $\alpha_n = q_n/p_n > 1$ for $n \geq 2$, and they differ because w_n^∞ is periodic while $1c_\gamma$ is not, γ being irrational. $\square$

In particular the threshold $n \geq 3$ is uniform in γ : no step above introduces a bound depending on the slope or on its partial quotients.

The Liouville step, uniformly in the slope. At $\gamma = \beta$ the proof of Theorem 6.1 used the coincidence $\alpha_\beta = \log_2 3$ to write $3^{p_n} = 2^{q_n + \alpha D_n}$ with $\alpha D_n < 0.2$, so that the archimedean size of the approximants was 2^{q_n} up to a bounded factor. For a general slope the two powers 2^{q_n} and 3^{p_n} part exponentially — their ratio is $2^{(\theta-1)q_n + O(1)}$ — and the point of the next theorem is that the depth law still outruns the larger of them, by a margin bounded below independently of γ .

Theorem 8.4. *Let $\gamma \in (0, 1)$ be irrational and suppose $\Phi(1c_\gamma) = u/v$ with $u \in \mathbb{Z}$, $v \geq 1$, $\gcd(u, v) = 1$; put $H = \max(|u|, v)$. Then every odd $n \geq 3$ satisfies*

$$q_{n+1} - \kappa q_n < \log_2 H + \log_2 q_n + 2 + \log_2 3 \quad (11)$$

and therefore

$$c(\gamma) q_n - \log_2 q_n < \log_2 H + 2 + \log_2 3. \quad (12)$$

Proof. As $\Phi(1c_\gamma) \in \mathbb{Z}_2$, the denominator v is odd. Fix an odd $n \geq 3$, put $\delta_n = 2^{q_n} - 3^{p_n}$, $c_n = c_{w_n}$, so that $\Phi(w_n^\infty) = c_n/\delta_n$ by Proposition 4.1, and set

$$G_n = \max(2^{q_n}, 3^{p_n}), \quad M_n = u \delta_n - v c_n \in \mathbb{Z}, \quad \Phi(1c_\gamma) - \Phi(w_n^\infty) = \frac{M_n}{v \delta_n}.$$

Lower bound. Both v and δ_n are odd, so $v_2(M_n) = v_2(\Phi(1c_\gamma) - \Phi(w_n^\infty)) = q_n + q_{n+1} - 1$ by Theorem 8.3. The valuation is finite, so $M_n \neq 0$ and

$$|M_n| \geq 2^{q_n + q_{n+1} - 1}. \quad (13)$$

Upper bound. The powers 2^{q_n} and 3^{p_n} are distinct positive integers, so $|\delta_n| < G_n$; and $c_n \leq q_n G_n$ by Lemma 4.2, which is stated for an arbitrary mechanical word and makes no assumption on which power is the larger. Hence

$$|M_n| \leq |u| |\delta_n| + v c_n < H(1 + q_n) G_n \leq 2H q_n G_n. \quad (14)$$

The size of G_n . Since n is odd, $p_n = \gamma q_n + D_n$ with $0 < D_n < 1/q_{n+1} \leq 1$, so

$$\log_2 G_n = \max(q_n, p_n \log_2 3) \leq \max(q_n, \theta q_n + \log_2 3) \leq \max(1, \theta) q_n + \log_2 3 = (1 + \kappa) q_n + \log_2 3.$$

Comparison. Combining (13) and (14),

$$q_n + q_{n+1} - 1 < 1 + \log_2 H + \log_2 q_n + \log_2 G_n \leq 1 + \log_2 H + \log_2 q_n + (1 + \kappa) q_n + \log_2 3,$$

and cancelling q_n gives (11). Finally $q_{n+1} > q_n$ for $n \geq 1$, so $q_{n+1} - \kappa q_n > (1 - \kappa) q_n = c(\gamma) q_n$, which turns (11) into (12). $\square$

Theorem 8.5. $\Phi(1c_\gamma) \notin \mathbb{Q}$ for every irrational $\gamma \in (0, 1)$. More precisely, with H as in Theorem 8.4, every odd $n \geq 3$ satisfies

$$q_n < 5 \log_2 H + 22, \quad (15)$$

which is impossible because $q_n \rightarrow \infty$ along the odd indices.

Proof. Put $c_0 = 2 - \log_2 3 = 0.41503\dots$, so that $c(\gamma) > c_0$ for every $\gamma \in (0, 1)$, because $\theta = \gamma \log_2 3 < \log_2 3$. Suppose some odd $n \geq 3$ had $q_n \geq 5 \log_2 H + 22$. Then $q_n \geq 22$, and the function $Q \mapsto Q/\log_2 Q$ attains its minimum at $Q = e$ and increases thereafter, so

$$\frac{q_n}{\log_2 q_n} \geq \frac{22}{\log_2 22} = 4.9334\dots > 4.8189\dots = \frac{2}{c_0},$$

whence $\log_2 q_n \leq \frac{c_0}{2} q_n$ and

$$c(\gamma)q_n - \log_2 q_n \geq c_0 q_n - \frac{c_0}{2} q_n = \frac{c_0}{2} q_n \geq \frac{c_0}{2} (5 \log_2 H + 22) = 1.0375 \dots \log_2 H + 4.5654 \dots$$

which exceeds $\log_2 H + 2 + \log_2 3 = \log_2 H + 3.5849 \dots$, contradicting (12). So no odd $n \geq 3$ has $q_n \geq 5 \log_2 H + 22$; but q_n is unbounded along the odd indices, and the contradiction is reached at the first odd n with $q_n \geq 5 \log_2 H + 22$. $\square$

Remark 8.6. Three remarks on the constants. First, for $\gamma \leq \beta$ one has $c(\gamma) = 1$ and the same computation gives the sharper $q_n < 2 \log_2 H + 8$. Second, at $\gamma = \beta$ one has $\kappa = 0$ and (11) reads $q_{n+1} < \log_2 H + \log_2 q_n + 3.585$, against $q_{n+1} \leq 3 + \log_2 H + \log_2 q_n$ in Theorem 6.1; the published constant is better only because the proof there knows that $q_4 = 8$ for β . Third, the margin in Theorem 8.5 is structural rather than numerical: the regime $3^{p_n} > 2^{q_n}$ costs the comparison a factor $2^{p_n \log_2 3 - q_n}$, which is at most $2^{(\theta-1)q_n + \log_2 3}$, and $\theta - 1 < \log_2 3 - 1 < 1 \leq q_{n+1}/q_n$ for every $\gamma \in (0, 1)$. The two rates would meet only at $\theta = 2$, that is at $\gamma = 2/\log_2 3 = 1.2618 \dots$, which is not a slope.

Corollary 8.7. *Let $\gamma \in (0, 1)$ be irrational and $n \geq 3$ odd. Then $\Phi(1c_\gamma)$ is not equal to any u/v in lowest terms with*

$$\max(|u|, v) \leq \frac{2^{q_n + q_{n+1} - 1}}{(1 + q_n) \max(2^{q_n}, 3^{p_n})}.$$

Proof. Immediate from (13) and (14). $\square$

For $\gamma = 1/\varphi = (\sqrt{5} - 1)/2$ and $n = 25$, where $p_n/q_n = 75025/121393$ and $q_{n+1} = 196418$, the bound excludes every height up to 2^{196400} ; for $\gamma = e - 2$ and $n = 13$, where $p_n/q_n = 12993/18089$ and $q_{n+1} = 190435$, it excludes every height up to 2^{187915} . These values, and the depths on which they rest, were computed in exact integer arithmetic.

Consequences.

Corollary 8.8. *Let $\gamma \in (0, 1)$ be irrational. Then $\Phi(c_\gamma)$, $\Phi(0c_\gamma)$ and $\Phi(\sigma^k 1c_\gamma)$ for every $k \geq 0$ are irrational. Moreover, for no odd $q \geq 1$ does the generalised map T_q possess an integer orbit whose parity word is $1c_\gamma$ or any shift of it.*

Proof. The proofs of Corollaries 7.1, 7.2 and 7.3 use only that $1c_\gamma$ begins with 1, that $0c_\gamma$ begins with 0, and that $\sigma(1c_\gamma) = \sigma(0c_\gamma) = c_\gamma$. The first two hold because $\lceil \gamma \rceil = 1$ and $\lfloor \gamma \rfloor = 0$ for $\gamma \in (0, 1)$; the third because for $j \geq 1$ and irrational γ , $\lceil (j+1)\gamma \rceil - \lfloor j\gamma \rfloor = (\lfloor (j+1)\gamma \rfloor + 1) - (\lfloor j\gamma \rfloor + 1) = \lfloor (j+1)\gamma \rfloor - \lfloor j\gamma \rfloor$. With Theorem 8.5 in place of Theorem 6.1, the three proofs go through unchanged. $\square$

Remark 8.9. *The depth law is not generalised to other intercepts. Everything in this section is at intercept 0; for the mechanical words of slope γ and intercept $\rho \neq 0$ the one-positions are no longer $\lfloor j\alpha_\gamma \rfloor$ and the exact identities in the proof of Lemma 5.2 — in particular $m_{p_{n+1}} = p_n - 1$ and $(p_n + p_{n+1})\alpha_\gamma D_n = 1 + p_n \alpha_\gamma (D_n - D_{n+1})$ — have no counterpart. The irrationality conclusion at $\rho \neq 0$ is nevertheless available: Section 10 obtains it for every intercept by a different route, at the cost of the weaker height floor recorded in Remark 10.6. What remains open is the quantitative statement, a depth law at general intercept; see Section 12 (1).*

9. A COUNTING BOUND, AFTER DUBICKAS

For $s \in \{0, 1\}^{\mathbb{N}}$ let $p_s(L)$ denote the number of distinct length- L factors of s , and put $\lambda = \log_2(3/2) = 0.58496 \dots$. The following argument is Dubickas's [9, Theorem 5]: he treats a sequence of positive integers $x_n \rightarrow \infty$, using the divisibility $2^m \mid x_n - x_s$ forced by equal

length- m parity blocks and the growth bound $x_n < (3/2)^n(x_0 + 1)$. We record its transport to every rational 2-adic integer, which needs only that T preserve an odd denominator.

Lemma 9.1 (fixed denominator). *Let $x = a/b$ with $\gcd(a, b) = 1$, $b \geq 1$ odd, and write $T^t(x) = a_t/b$. Then $a_t \in \mathbb{Z}$ and $|a_t| + b \leq (3/2)^t(|a| + b)$ for all $t \geq 0$.*

Proof. Since $x \in \mathbb{Z}_2$, b is automatically odd: if b were even then, as $\gcd(a, b) = 1$, a would be odd and $v_2(a/b) = -v_2(b) < 0$. The parity of a_t/b in $\mathbb{Z}_2$ is therefore that of a_t . If a_t is even then $a_{t+1} = a_t/2$; if a_t is odd then $a_{t+1} = (3a_t + b)/2 \in \mathbb{Z}$, since $3a_t + b$ is then even. In both cases $|a_{t+1}| + b \leq (3|a_t| + b)/2 + b = \frac{3}{2}(|a_t| + b)$; induct on t . $\square$

Proposition 9.2. *Let $s \in \{0, 1\}^{\mathbb{N}}$ be not eventually periodic and suppose $\Phi(s) = a/b$ in lowest terms; put $H_0 = |a| + b$. Then for every $L \geq 1$,*

$$p_s(L) \geq \#\{t \geq 0 : 3^t H_0 \leq 2^{L-1+t}\} = \left\lfloor \frac{L-1-\log_2 H_0}{\lambda} \right\rfloor + 1 \quad (L-1 \geq \log_2 H_0).$$

Proof. Write $x_t = T^t(\Phi(s)) = a_t/b$ as in Lemma 9.1. If $x_i = x_j$ for some $i < j$ then $\sigma^i s = \sigma^j s$ by Proposition 2.3 and bijectivity of Φ , so s is eventually periodic, contrary to hypothesis; hence the a_t are pairwise distinct.

The factor $s_t \cdots s_{t+L-1}$ is the length- L parity prefix of x_t , hence determines $x_t \bmod 2^L$ by Proposition 2.2, hence $a_t \bmod 2^L$, since b is a unit mod 2^L . For each t with $(3/2)^t H_0 \leq 2^{L-1}$, Lemma 9.1 gives $|a_t| \leq 2^{L-1} - b < 2^{L-1}$, and distinct integers in $(-2^{L-1}, 2^{L-1})$ are distinct modulo 2^L ; so the factors at these times are pairwise distinct. Since $(3/2)^t H_0$ is increasing in t , the admissible t form the initial segment $0 \leq t \leq \lfloor (L-1-\log_2 H_0)/\lambda \rfloor$, of size $\lfloor (L-1-\log_2 H_0)/\lambda \rfloor + 1$. $\square$

Corollary 9.3. *If s is not eventually periodic and $\liminf_{L \rightarrow \infty} p_s(L)/L < 1/\lambda = 1.70951\dots$, then $\Phi(s) \notin \mathbb{Q}$.*

Proof. Dividing Proposition 9.2 by L gives $p_s(L)/L \geq 1/\lambda - (1 + \log_2 H_0)/(\lambda L)$ once $L-1 \geq \log_2 H_0$; so if $\Phi(s) = a/b \in \mathbb{Q}$ then $\liminf_L p_s(L)/L \geq 1/\lambda$. The contrapositive is the claim. $\square$

Corollary 9.4 (Sturmian, effective). *If s is Sturmian, so $p_s(L) = L + 1$ for every $L \geq 1$, then $\Phi(s) \notin \mathbb{Q}$. Quantitatively, no rational a/b with $H_0 = |a| + b$ equals $\Phi(s)$ once*

$$L > \frac{\log_2 H_0 + 1 + \lambda}{2 - \log_2 3}.$$

Proof. A Sturmian word is not eventually periodic. If $\Phi(s) = a/b \in \mathbb{Q}$, Proposition 9.2 with $p_s(L) = L + 1$ gives $L + 1 \geq (L-1-\log_2 H_0)/\lambda$ for every L with $L-1 \geq \log_2 H_0$, i.e. $(1-\lambda)L \leq \log_2 H_0 + 1 + \lambda$; since $1-\lambda = 2 - \log_2 3 = 0.41503\dots > 0$, this fails once $L > (\log_2 H_0 + 1 + \lambda)/(2 - \log_2 3)$, a contradiction. $\square$

Remark 9.5. The margin $2 - \log_2 3$ in Corollary 9.4 is the worst-case margin $c(\gamma)$ of Theorem 10.1 below: the two proofs trade repetition against height at the same rate. The counting route covers every word s that is not eventually periodic and satisfies $p_s(L) \leq cL + O(1)$ for some $c < 1/\lambda = 1.70951\dots$ (Corollary 9.3), quasi-Sturmian words included; the Liouville route of Sections 6–10 gives in addition the exact depth law (Theorem 5.3) and height floors along the convergents (Corollary 6.2, Corollary 8.7). Neither route reaches complexity $2L$ (Rote words, binary codings of rotations with two intervals); see Section 12, problem (4).

The exact-arithmetic checks for this section are recorded in Appendix A, table `dubickas-counting`; the accompanying script is `scripts/dubickas_counting_check.py`.

10. ALL STURMIAN WORDS

Section 8 settles every irrational slope at intercept 0. The restriction to intercept 0 is forced by the method: the depth law rests on two exact identities that hold only when the ones sit at $\lfloor j\alpha_\gamma \rfloor$. This section removes the restriction by a different route, which uses no depth law at all. Instead of comparing the word with the periodic words attached to the convergents of its slope, we compare it with the periodic word attached to an *initial square* of the word itself.

For irrational $\gamma \in (0, 1)$ and $\rho \in [0, 1)$ the two mechanical words of slope γ and intercept ρ are

$$s_{\gamma, \rho}(j) = \lfloor (j+1)\gamma + \rho \rfloor - \lfloor j\gamma + \rho \rfloor \quad \text{and} \quad s'_{\gamma, \rho}(j) = \lceil (j+1)\gamma + \rho \rceil - \lceil j\gamma + \rho \rceil$$

for $j \geq 0$; both are Sturmian, every Sturmian word arises this way, and $1c_\gamma = s'_{\gamma, 0}$, $0c_\gamma = s_{\gamma, 0}$. We write X_γ for the Sturmian subshift of slope γ , the set of all of them.

Theorem 10.1. *Let $\gamma \in (0, 1)$ be irrational, let $\rho \in [0, 1)$, and let s be either mechanical word of slope γ and intercept ρ . Then $\Phi(s) \notin \mathbb{Q}$. More precisely, if $\Phi(s) = u/v$ in lowest terms with $H = \max(|u|, v)$, then every half-length ℓ of an initial square of s satisfies*

$$c(\gamma)\ell \leq \log_2 H + \log_2(2 + 3\ell) + \log_2 3, \quad (16)$$

where $c(\gamma) = 2 - \max(1, \gamma \log_2 3) \geq 2 - \log_2 3 = 0.41503\dots$, and such ℓ are unbounded.

The proof needs two facts about initial repetitions in Sturmian words. Both are quoted; neither is ours.

Theorem 10.2 (Allouche–Davison–Queff  lec–Zamboni). *Every Sturmian word begins in infinitely many squares. Consequently its initial critical exponent — the supremum of the real $r > 0$ for which arbitrarily long prefixes u have u^r a prefix — is at least 2.*

This is [1, Theorem 1]; see also [6], and [4, §1], where it is quoted in this form and where the exact value of the initial critical exponent is computed from the Ostrowski expansion of the intercept. There are *no exceptional slopes or intercepts*. Since a word has at most one prefix of each length, “infinitely many squares” gives infinitely many distinct half-lengths, so the half-lengths are unbounded, which is what (16) needs. The exponent 2 cannot be improved in general: [4, Theorem 1.1] characterises the slopes admitting a Sturmian word of initial critical exponent exactly 2. It is enough here because $2 > \max(1, \gamma \log_2 3)$ for every $\gamma \in (0, 1)$, with the margin $c(\gamma)$ of (16).

Theorem 10.3 (Berth  –Holton–Zamboni). *If a Sturmian word begins in w^r with $r \geq 2$, $|w| \geq 2$ and w primitive, then w is a cyclic permutation of a standard word. In particular w^∞ is balanced.*

This is [4, Proposition 3.2], which produces an $m \geq 0$ with w a cyclic permutation of $\tau_{i_1} \circ \dots \circ \tau_{i_m}(01)$, where $\tau_0 : 0 \mapsto 0, 1 \mapsto 01$ and $\tau_1 : 0 \mapsto 10, 1 \mapsto 1$. Applying these morphisms to the standard pair $(0, 1)$ yields standard pairs, and the concatenation of a standard pair is a standard word [13, Ch. 2]. If w is not primitive, $w = v^t$ and the word begins in v^{2t} , so primitivity costs nothing.

The hypothesis of Theorem 10.3 is doing real work, and it is not implied by balance of w alone: the balanced words 0110, 1001, 01110 and 10001 have unbalanced periodizations, and none is a cyclic permutation of a standard word. What Lemma 10.4 needs is balance of w^∞ , which is what Theorem 10.3 supplies.

Lemma 10.4. *Let W be a cyclic permutation of a standard word, $|W| = \ell$, with $k \geq 1$ ones. Then $0 < c_W \leq 3\ell \max(2^\ell, 3^k)$.*

Proof. Positivity is clear from (2). By Theorem 10.3 the word W^∞ is balanced, so $|k_{i+1}(W) - (i+1)k/\ell| < 1$ and hence $k - k_{i+1}(W) < k(\ell - 1 - i)/\ell + 1$. With $\rho = 3^{k/\ell}$,

$$3^{k-k_{i+1}(W)} 2^i < 3 \rho^{\ell-1-i} 2^i \leq 3 \max(\rho^{\ell-1}, 2^{\ell-1}) \leq 3 \max(3^k, 2^\ell),$$

the middle step because $i \mapsto \rho^{\ell-1-i} 2^i$ is monotone, so its maximum over $0 \leq i \leq \ell - 1$ is attained at an endpoint. The sum (2) has at most ℓ terms. $\square$

This is the estimate sketched at the end of Remark 4.3, applied where it is needed. The constant 3 is not optimal: over every cyclic permutation of every standard word of length at most 40 the true value of $c_W/(\ell \max(2^\ell, 3^k))$ never exceeds 0.4394, attained at slope 12/19 (Appendix A), so $C = 1$ appears to suffice. Nothing below depends on the constant.

Proof of Theorem 10.1. Suppose $\Phi(s) = u/v$ in lowest terms and put $H = \max(|u|, v)$. Since $\Phi(s) \in \mathbb{Z}_2$, the denominator v is odd. Let WW be an initial square of s with W primitive, $|W| = \ell$ and k ones, and set $\delta = 2^\ell - 3^k$, which is odd because $\ell \geq 1$. Put

$$M = u\delta - v c_W \in \mathbb{Z}.$$

If $M = 0$ then $\Phi(s) = u/v = c_W/\delta = \Phi(W^\infty)$ by Proposition 4.1, and Φ is a bijection, so $s = W^\infty$. That contradicts the aperiodicity of s , which holds because γ is irrational. Hence $M \neq 0$ and $|M| \geq 1$.

Both s and W^∞ begin with WW , so $\text{lcp}(s, W^\infty) \geq 2\ell$. Now $\Phi(s) - \Phi(W^\infty) = M/(v\delta)$ with v and δ odd, so $v_2(M) = v_2(\Phi(s) - \Phi(W^\infty))$, which is $\text{lcp}(s, W^\infty) \geq 2\ell$ by Proposition 2.2. Since M is a nonzero integer, $|M| \geq 2^{2\ell}$.

In the other direction W is a cyclic permutation of a standard word by Theorem 10.3, so Lemma 10.4 gives

$$|M| \leq H(|\delta| + c_W) \leq H(2^\ell + 3^k + 3\ell \max(2^\ell, 3^k)) \leq H(2 + 3\ell) \max(2^\ell, 3^k),$$

whence $2\ell \leq \log_2 H + \log_2(2 + 3\ell) + \max(\ell, k \log_2 3)$. Finally W is a factor of a Sturmian word of slope γ , so $|k - \gamma\ell| \leq 1$ and $\max(\ell, k \log_2 3) \leq \max(1, \gamma \log_2 3) \ell + \log_2 3$. Rearranging gives (16). By Theorem 10.2 there are such ℓ arbitrarily large, while $c(\gamma) \geq 2 - \log_2 3 > 0$ because $\gamma < 1$; so (16) fails for all large ℓ , and no such u/v exists. $\square$

Corollary 10.5. *For every irrational $\gamma \in (0, 1)$, $\Phi(X_\gamma) \cap \mathbb{Q} = \emptyset$.*

Proof. X_γ is exactly the set of mechanical words of slope γ over all intercepts, so this is Theorem 10.1 restated. $\square$

Theorem 6.1 and Theorem 8.5 are the case $\rho = 0$, and Corollary 7.2 the case $\rho \in \{\{j\gamma\} : j \geq 0\}$; those are countably many words per slope, while X_γ is uncountable. Corollary 8.8 extends verbatim: for no odd $q \geq 1$ does the generalised map T_q possess an integer orbit whose parity word is any Sturmian word.

Remark 10.6. The two routes are independent and their strengths differ. The depth law of Section 8 exploits the continued fraction of γ and gives, at intercept 0, the height floor of Corollary 8.7, of order $2^{q_n + q_{n+1} - 1}$. Theorem 10.1 gives, at every intercept, a floor of order $2^{c(\gamma)\ell}$ with ℓ running over the half-lengths of the initial squares. At intercept 0 those half-lengths are the convergent and semiconvergent denominators of γ , and the first bound is the stronger of the two; away from intercept 0 only the second is available.

Remark 10.7. Two different quantities are written q in this paper and should not be confused. In Corollary 7.3 and Corollary 8.8 the letter q is the *additive* constant of the map $T_q(n) =$

$(3n + q)/2$, whose multiplier is still 3; no logarithm of q enters, and the range “ $q \geq 1$ odd” is unaffected by anything in this section. Elsewhere q_n denotes a convergent denominator.

It is worth recording what would change for a different *multiplier*. Nothing in this paper treats one, but if the argument of Theorem 10.1 were run for the conjugacy map of a $qx + 1$ map with q odd, the margin in (16) would become $2 - \max(1, \gamma \log_2 q)$, which is positive for every $\gamma \in (0, 1)$ if and only if $\log_2 q \leq 2$, that is only for $q = 3$. For $q \geq 5$ it vanishes at $\gamma = 2/\log_2 q < 1$ and is negative above it, so the initial-square route would fail at the large slopes. The same threshold appears in the last sentence of the remark following Theorem 8.5, where the two rates meet at $\theta = 2$, that is at $\gamma = 2/\log_2 3 = 1.2618\dots$, which is not a slope *because the multiplier is 3*.

11. WHY THE 2-ADIC PLACE

The series of Proposition 3.2 has no archimedean meaning at this slope, and this is not an accident of presentation. Writing $\lfloor j\alpha \rfloor = j\alpha - \{j\alpha\}$ and $2^\alpha = 3$,

$$3^{-(j+1)} 2^{\lfloor j\alpha \rfloor} = \frac{1}{3} 2^{-\{j\alpha\}} \in \left(\frac{1}{6}, \frac{1}{3}\right] \quad \text{for every } j \geq 0,$$

so the terms do not tend to 0 and the real series diverges. López and Stoll reach the same point from the other side: their real staircase diverges at $x = \beta$, and they call the object at that slope a divergent series.

The same identity $2^\alpha = 3$ places the evaluation point on the boundary of the domain of every relevant archimedean theorem. The Hecke–Mahler series $f(z_1, z_2) = \sum_{j \geq 0} z_1^j z_2^{\lfloor j\alpha \rfloor}$ is being evaluated at $(z_1, z_2) = (1/3, 2)$, where $|z_1 z_2^\alpha| = \frac{1}{3} \cdot 3 = 1$ exactly. Bugeaud and Laurent [5] prove transcendence at every algebraic point with $0 < |z_1|, |z_1 z_2^\theta| < 1$ *strictly*, and give explicit irrationality exponents at rational points strictly inside that domain; our point lies on its boundary. The results of Luca, Ouaknine and Worrell [14, 15] concern series $\sum_n f(\lfloor n\theta + \rho \rfloor) \gamma^{-n}$ with f a polynomial, so with coefficients of polynomial growth; here the coefficients $2^{\lfloor j\alpha \rfloor}$ grow like 3^j , which is precisely what makes the terms $\Theta(1)$.

At the 2-adic place the same resonance is harmless: $v_2(2^{\lfloor j\alpha \rfloor}) = \lfloor j\alpha \rfloor \rightarrow \infty$ while $3^{-(j+1)}$ is a unit, so the series converges, and the approximation problem becomes the comparison carried out in Section 6. The failure of the archimedean theory here is a relocation, not a degeneracy.

It is worth contrasting the situation with the rational base $3/2$. For the minimal word of that base, Stephan [18] obtains transcendence criteria, but the irrationality of the associated constant remains open even in the second version of that preprint (2026-09-17), which adds a 2-adic rigidity lemma for that single orbit (its Theorem 4.2, the mechanism behind Lemma 9.1 and Proposition 9.2 here) and shows in its §5.2 that rationality of the orbit’s limiting constant would force superlinear complexity, conditionally settling that one case; the general statement remains open there, and the natural place for it is archimedean. Here it is the 2-adic place at which the argument closes, both in Section 9 and in the rest of the paper, and the reason is the isometry of Proposition 2.2, which converts the approximation depth into a longest common prefix and so into the combinatorics of Sturmian words [13].

12. OPEN PROBLEMS

- (1) *A depth law at general intercept.* The qualitative question asked here in an earlier version — is $\Phi(1c_{\gamma, \rho})$ irrational for every irrational γ and every ρ ? — is *settled* by Theorem 10.1. What remains open is the quantitative statement. Let $1c_{\gamma, \rho}(j) = \lfloor (j+1)\gamma + \rho \rfloor - \lfloor j\gamma + \rho \rfloor$ and let w_n be the periodic word attached to the convergent p_n/q_n at a matched intercept. Is there an explicit $L_n(\gamma, \rho)$ with $\text{lcp}(1c_{\gamma, \rho}, w_n^\infty) = L_n(\gamma, \rho)$

and $L_n \geq q_n + q_{n+1} - O(1)$ for infinitely many n , uniformly in ρ ? The obstruction is precise. The ones of $1c_{\gamma,\rho}$ no longer sit at $\lfloor j\alpha_\gamma \rfloor$, so the quantity $m_j = jq_n \bmod p_n$ of Lemma 5.2 acquires a shift, and with it the two exact identities on which that proof turns — $m_{p_{n+1}} = p_n - 1$ from $p_{n+1}q_n \equiv -1 \pmod{p_n}$, and $(p_n + p_{n+1})\alpha_\gamma D_n = 1 + p_n\alpha_\gamma(D_n - D_{n+1})$ — lose their right-hand sides, and the first mismatch index j_n^* is no longer $p_n + p_{n+1}$. The natural replacement is the first entry time of the orbit $\{j\gamma + \rho\}$ into an interval of length jD_n at the partition endpoint, which the three-distance theorem and the Ostrowski expansion of ρ compute; [4] carries out exactly that kind of computation for the initial critical exponent. The height bound (Lemma 4.2) and the Liouville comparison (Theorem 8.4) survive unchanged, so such a depth law would restore at every intercept the sharper height floor of Corollary 8.7, which Theorem 10.1 does not give.

- (2) *An irrationality measure.* Is there $\mu < \infty$ and $C > 0$ with $|\Phi(1c_\gamma) - y|_2 \geq C|y|^{-\mu}$ for all integers $y \geq 1$, for a given irrational $\gamma \in (0, 1)$? Theorem 8.4 bounds the height of a rational *equal* to $\Phi(1c_\gamma)$, which is the case y exactly realising the value; a measure would bound how well every rational approximates it, and the argument here gives nothing of the sort, because it uses only the approximants $\Phi(w_n^\infty)$. By [8, Theorem 5.2] any finite μ at $\gamma = \beta$ would give an exponential lower bound on the least positive integers realising the prefixes of this word.
- (3) *Transcendence.* Is $\Phi(1c_\beta)$, or $\Phi(1c_\gamma)$ for a given irrational γ , transcendental over $\mathbb{Q}$? The archimedean analogue is known for the Hecke–Mahler values strictly inside the domain [5]; here one would want a 2-adic Mahler method along the continued-fraction chain.
- (4) *Complexity between the counting threshold and $2L$.* Corollary 9.3 reaches every s with $\liminf_L p_s(L)/L < 1/\log_2(3/2) = 1.70951\dots$; Rote words and binary codings of rotations with two intervals sit at $p_s(L) = 2L$, out of reach of counting alone (Remark 9.5). Is $\Phi(s) \notin \mathbb{Q}$ for every non-eventually-periodic s with $p_s(L) \leq 2L$ for all large L — in particular for Rote words? The repetition method of Sections 6–10 needs, in place of Theorem 10.2 and Theorem 10.3, some substitute initial repetition and height bound valid on this wider class; none is known to the author. This is the natural next target for either method.

None of these, and none of Theorem 6.1, Theorem 8.5 and Theorem 10.1, bears on divergent orbits of positive integers, on the Periodicity Conjecture in general, or on the Collatz conjecture. The Periodicity Conjecture quantifies over all aperiodic parity vectors; the Sturmian words form a two-parameter family among them — uncountable, but of measure zero in the space of all parity vectors — and outside the critical density the conclusion for that family was already available ([16, Theorem 2.7(b)] below β , [12, Theorem 1] above it).

APPENDIX A. VERIFICATION

The script `scripts/verify.py` accompanying this paper carries out the computations below in exact integer and 2-adic arithmetic; no floating-point value enters any decision. Each command is named with the claim it checks.

- **convergents** — Convention 5.1: the continued fraction of β to 368 rigorously determined partial quotients, the convergents p_n/q_n for $n \leq 14$, and the side of each, decided by the exact integer comparison 3^{p_n} against 2^{q_n} .
- **lcp** — Theorem 5.3 computed as a longest common prefix of the two 0/1 words.

- **depths** — Theorem 5.3 computed instead 2-adically, as v_2 of the difference of $\Phi(1c_\beta)$ and c_n/δ_n modulo a power of 2 exceeding the predicted depth. This is independent of the previous item, and agreeing values confirm Proposition 2.2 as well.
- **heights** — Lemma 4.2 for every mechanical word of slope p/q with $q \leq 120$, and the closed form $c_w = 2^a(3^a - 2^a)$ of Remark 4.3.
- **controls** — the two rational controls described below.
- **shadow** — the explicit height of Corollary 6.2.
- **slopes** — Theorem 8.3 at seven irrational slopes, computed twice and independently, as a longest common prefix and 2-adically, as above.
- **uniform** — every explicit constant of Section 8; the step $\log_2 G_n \leq \max(1, \theta)q_n + \log_2 3$ of Theorem 8.4 in its exact form $p_n - \gamma q_n \leq 1$, at every odd convergent of every test slope; and the exact height floor of Corollary 8.7 at each of them.
- **slopecontrols** — the rational-slope control for Section 8, described below.
- **dubickas-counting** — Lemma 9.1 and Proposition 9.2, checked by the separate, independent script `scripts/dubickas_counting_check.py`, described below.
- **squares** — Theorem 10.2 at four slopes (β , $1/\phi$, $\sqrt{2} - 1$ and $[0; 1, 97, 1, \dots]$) and six intercepts each, including intercepts not of the form $\{j\gamma\}$, and for both the lower and the upper mechanical word.
- **squares-random** — the same over many random intercepts per slope; it reports how many produce no initial square at all.
- **roots** — Theorem 10.3: each initial-square root is checked to be a cyclic permutation of the standard word of its own slope, by exact set membership among all ℓ rotations.
- **conjheight** — Lemma 10.4 over *every* cyclic permutation of *every* standard word of length at most 40; it reports the true maximum of $c_W/(\ell \max(2^\ell, 3^k))$.
- **wordskey** — the surplus $2\ell - \max(\ell, k \log_2 3)$ of (16) at each square found, against $c(\gamma)\ell$, together with the discrepancy bound $|k - \gamma\ell| \leq 1$.
- **wordcontrols** — the controls for Section 10, described below.

Depths. Theorem 5.3 was verified at every level $3 \leq n \leq 14$, that is, out to the convergent 190537/301994. Each entry was computed twice and independently: as a longest common prefix of the two 0/1 words, and 2-adically as $v_2(\Phi(1c_\beta) - c_n/\delta_n)$ modulo 2^{L+40} , where L is the predicted depth; the deepest modulus used is 2^{478284} . All twenty-four values agree with the theorem.

n	p_n/q_n	predicted depth	computed
3	2/3	$q_3 + q_4 - 1 = 10$	10
4	5/8	$q_4 - 1 = 7$	7
5	12/19	$q_5 + q_6 - 1 = 83$	83
6	41/65	$q_6 - 1 = 64$	64
7	53/84	$q_7 + q_8 - 1 = 568$	568
8	306/485	$q_8 - 1 = 484$	484
9	665/1054	$q_9 + q_{10} - 1 = 25780$	25780
10	15601/24727	$q_{10} - 1 = 24726$	24726
11	31867/50508	$q_{11} + q_{12} - 1 = 176250$	176250
12	79335/125743	$q_{12} - 1 = 125742$	125742
13	111202/176251	$q_{13} + q_{14} - 1 = 478244$	478244
14	190537/301994	$q_{14} - 1 = 301993$	301993

Heights. Lemma 4.2 holds with no violation over every mechanical word $w_{p,q}$ with $q \leq 120$; the largest observed value of $c_w/(q \max(2^q, 3^p))$ is $1/3$, attained at $p/q = 1/1$.

Controls. A positive verification of a depth law is only meaningful if the same computation fails to produce a contradiction when run against a rational target, since the conclusion of Theorem 6.1 must not be derivable there. Two controls are computed. Taking as target the rational 2-adic integer $\Phi(w_3^\infty) = -5$, the depths $v_2(-5 - \Phi(w_n^\infty))$ do not grow: they freeze at $10 = q_3 + q_4 - 1$. Taking instead $\Phi(w_5^\infty)$, they freeze at $83 = q_5 + q_6 - 1$. In each case the agreement depth is bounded while the archimedean height of the approximants grows linearly, so the comparison of Section 6 yields nothing — exactly as it must not, these targets being rational.

General slopes. Theorem 8.3 was verified at seven irrational slopes, chosen to stress the argument in different ways: the two quadratic irrationals $1/\varphi$ and $\sqrt{2} - 1$; the transcendental $e - 2$; $[0; 50, 1, 1, \dots]$, whose $\alpha_\gamma = 50.61\dots$ is large, so that Lemma 8.2 is doing work; $[0; 1, 97, 1, \dots]$, with a large early partial quotient and $\gamma = 0.98986$, the case closest to the edge $c(\gamma) \rightarrow 2 - \log_2 3$; $[0; 1, 1, 1, 1000, 1, \dots]$, with a large later partial quotient; and $\gamma \approx 1/(7 + 1/\pi) = 0.13664$. Three lie below β and four above, and none is β . Every $\lfloor j\alpha_\gamma \rfloor$ is certified by bracketing α_γ between two consecutive convergents $q_m/p_m, q_{m+1}/p_{m+1}$ and requiring the two floors to agree.

At each slope, at every level n with predicted depth below the cap, the depth and the first mismatch index j_n^* were computed twice — as a longest common prefix and 2-adically — and compared with Theorem 8.3. All values agree, with no exception at any level and no failure at small n : in fact the law was observed to hold already from $n = 1$, although the proof is given only for $n \geq 3$. The table records the deepest odd level reached at each slope, the regime there (A: $2^{q_n} \geq 3^{p_n}$; B: the reverse), and the exact height floor of Corollary 8.7.

slope γ	continued fraction	n	q_n	q_{n+1}	regime	$\log_2$ of the height floor
$1/\varphi$	$[0; 1, 1, 1, \dots]$	25	121393	196418	A	196400
$\sqrt{2} - 1$	$[0; 2, 2, 2, \dots]$	13	80782	195025	A	195007
$e - 2$	$[0; 1, 2, 1, 1, 4, \dots]$	13	18089	190435	B	187915
$0.019755\dots$	$[0; 50, 1, 1, \dots]$	17	80837	130797	A	130779
$0.989859\dots$	$[0; 1, 97, 1, \dots]$	17	97336	157493	B	102101
$0.666555\dots$	$[0; 1, 1, 1, 1000, \dots]$	11	63081	102131	B	98552
$0.136643\dots$	$[0; 7, 3, 7, 15, \dots]$	3	161	2437	A	2428

The regime switch predicted in Section 8 is visible: for $\gamma = 1/\varphi = 0.61803\dots < \beta$ the odd convergent at $n = 3$ is $2/3 = 0.6667 > \beta$ and lies in regime B, while those from $n = 5$ on ($5/8, 13/21, \dots$) lie below β and in regime A. Theorem 8.4 never branches on the regime, so the switch costs nothing. The slope $[0; 1, 97, 1, \dots]$ shows the toll of regime B directly: its height floor is 2^{102101} where a regime A slope with the same q_n, q_{n+1} would give about 2^{157480} , and the margin $c(\gamma) = 0.4311$ is still comfortably positive.

Rational slopes. The control for Section 8 is the same in spirit. For rational p/q one has $1c_{p/q} = w_{p,q}^\infty$ exactly — verified over $8q$ letters for $p/q = 5/8, 2/3, 41/65$ — so $\Phi(1c_{p/q}) \in \mathbb{Q}$ and no contradiction may be derivable. At $\gamma = 5/8 = [0; 1, 1, 1, 2]$, where $\Phi(1c_\gamma) = 319/13$ and $H = 319$, the argument stops in two independent ways, both computed: the odd levels that exist give a satisfied bound rather than a violated one ($n = 3$ forces only $H > 28$), and there is no deeper level, because $n = 4$ is the last convergent and is even, the two words coincide there, and $M_4 = u\delta_4 - vc_4 = 319 \cdot 13 - 13 \cdot 319 = 0$ exactly, so the lower bound $|M_n| \geq 2^{q_n + q_{n+1} - 1}$ is vacuous. Equivalently, $q_n \rightarrow \infty$ along odd n fails, the continued fraction of $5/8$ being finite.

Dubickas counting bound (Section 9). The script `scripts/dubickas_counting_check.py`, independent of `verify.py`, checks Lemma 9.1 and Proposition 9.2 in exact integer and `Fraction`

arithmetic; no floating-point value enters any decision, floats appearing only in printed summaries. Running it prints **ALL CHECKS PASS**.

check	result
Lemma 9.1 growth, 51 rationals ($a \in \{\pm 1, \pm 5, 7, \pm 27, 319, 1001, -4097\}$, $b \in \{1, 3, 5, 7, 11, 13\}$, 60 steps)	0 violations
Lemma 9.1 rigidity ($a_t \bmod 2^L$ from the length- L factor), $L = 1, \dots, 14$	0 violations
Window distinctness (Proposition 9.2), 526 (rational, L) windows	0 violations
$1c_\beta$ prefix of length 20 000: $p(L) = L + 1$ for $L = 1, \dots, 40$	0 violations
First excluding L for $H_0 = 2^{10}, 2^{100}, 2^{1000}$ (formula $(\log_2 H_0 + 1 + \lambda)/(2 - \log_2 3)$)	28, 245, 2414 (27.9, 244.8, 2413.2)
Control: periodic 10, 11010, 10110101, true Φ values 1, 23/5, 485/13	inequality fails, as it must
Control: $x = -5$ (a 3-cycle of T)	repeats at $t = 3$, as required

The controls are the same discipline as elsewhere in this appendix: a positive check of Proposition 9.2 is only meaningful if the same computation fails on a target where the hypothesis is false. The three periodic words and the cyclic point $x = -5$ all violate the counting inequality at a finite L , exactly because they are eventually periodic and the hypothesis of Proposition 9.2 excludes them.

All Sturmian words. Slopes and intercepts are carried as integers over 2^{512} , so every floor is an integer shift; the script reports the minimum distance of $j\gamma + \rho$ to $\mathbb{Z}$ over the range used, in units of 2^{-512} . That margin was at least 496 bits in every run, against a representation error of at most $j \leq 2^{13}$ units, which certifies every floor.

Initial squares were found at every one of the 48 slope/intercept/parity combinations of **squares**, and at every one of 1800 random intercepts across three slopes in **squares-random**, with none failing; the smallest “largest half-length below 1200” observed was 169. The half-lengths are the convergent and semiconvergent denominators of the slope. All 48 initial-square roots tested by **roots** are cyclic permutations of standard words, with no exception. Over every cyclic permutation of every standard word of length at most 40, **conjheight** finds $c_W/(\ell \max(2^\ell, 3^k)) \leq 0.439399$, attained at slope 12/19 with $W = 0101101101011011011$; the bound of Lemma 10.4 with $C = 3$ is never approached. In **wordskey** the surplus $2\ell - \max(\ell, k \log_2 3)$ agreed with $c(\gamma)\ell$ to three decimals at every square, and $|k - \gamma\ell| \leq 1$ held throughout.

Controls for Section 10. Three, all of which must leave the argument silent. At a rational slope the word is periodic, so Φ of it is rational and $M = u\delta - vc_W = 0$ exactly; this was checked at $p/q = 2/3, 5/8, 12/19, 41/65$ in exact rational arithmetic, with the periodicity verified over eight periods. The Thue–Morse word is aperiodic but not Sturmian, and it begins in *no* square at all below half-length 2048, so Theorem 10.2 has no input and the method never starts. Finally, for $W = 0^a 1^a$ — balanced, but with unbalanced periodization — the ratio $c_W/(\ell \max(2^\ell, 3^k))$ takes the values 0.87, 2.83, 10.39, 41.02 at $a = 6, 10, 14, 18$ and diverges, confirming that Lemma 10.4 cannot be extended past Theorem 10.3.

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